

Residue-driven constituent curvature and the original arithmetic Laplacian

Exact rank-one transport, a source-sensitive curvature coefficient, and the retained period correction

Research continuation

13 September 2026

1 Scope and source identifications

This note continues the supplied *Deligne Split Cohomology* sidebar (DS1–69), *Exponential Period Determinant* (XD1–26), *Exponential Translation Comparison* (DT1–13), and the pasted SGA trace/period continuation. It uses their explicit coefficient and period maps. The additional results proved here concern the rank-one deformation on every nonzero proper invariant constituent, its exact period curvature, the change of that coefficient between the original source metrics, and the second-order realization of the arithmetic Laplacian. These are written proofs, supplemented by the included exact finite checker. They are not new Lean certificates or a proof of a uniform arithmetic endpoint bound.

Keep the original nonempty cyclic object, including its full multiplicities,

$$E = \mathbb{C}[S]/(\chi), \quad \chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad A = M_S. \quad (\text{RC1})$$

The coordinate basis is $1, S, \dots, S^{q-1}$. When $E = E_{h,k}$ is the arithmetic packet, the source retains $g = 2\xi$, $\mathcal{M}F_h = g/h$, the full arithmetic Taylor unit in η , and the original source/relation norm. In particular, retain the specified maps on the admitted strong-Schwartz tensor domain $\mathcal{X}$:

$$r_N : E \longrightarrow \mathcal{X}, \quad j_E : \mathcal{X} \longrightarrow E, \quad j_E r_N = I_E, \quad j_E D^{(k)} = A j_E, \quad j_E d = 0, \quad G_N = r_N^* r_N \succ 0. \quad (\text{RC2})$$

Here d is the original cochain differential and $D^{(k)}$ the original sum of scaling generators. Equation (RC2) is the supplied source comparison, not an assertion that the full-jet observation extends boundedly to the unweighted Hilbert completion (see *Exponential Intake*, §§4–5).

There are three distinct parameters. S is the arithmetic spectral coordinate; z is the earlier representative-curve parameter; t below is the exponential deformation parameter. The metric curve of DS13–14 is $G(z) = G_j + |z|^2(G_i - G_j)$. None of its radial derivatives is substituted for an exponential t -derivative.

For fixed $u \in \mathbb{C}^\times$, the supplied polynomial complex is

$$\mathcal{C}_{u,t} = [\mathbb{C}[S] \xrightarrow{\mathcal{D}_{u,t}} \mathbb{C}[S]dS], \quad \mathcal{D}_{u,t}P = (uP' + (\chi - t)P)dS. \quad (\text{RC3})$$

Its degree-one cohomology is a coefficient-vector-space quotient with the specified degree-below- q frame. Multiplication by S does not descend as an algebra operation on this general fibre: the supplied identities $[\chi - t] = 0$ and $[S(\chi - t)] = -u[1]$ rule out that interpretation. The parameter connection does descend, and in the retained frame it is

$$\nabla_t = \partial_t - \frac{A(t)}{u}, \quad A(t) = A + t\mathcal{R}, \quad \mathcal{R} = e_0 e_{q-1}^T. \quad (\text{RC4})$$

At $u = t = 0$ the complex specializes to the original quotient by χ . The regular operator $-u\nabla_t$ induces A on that special fibre; this is the type-correct specialization in DS68.

The fixed oriented contours of XD2 give a period matrix $\Pi(t) = \Pi(u, \mathbf{c}, t)$ satisfying

$$\Pi'(t) = -\frac{1}{u}\Pi(t)A(t), \quad \det \Pi(t) = \det \Pi_{\text{mon}}(u) \exp\left(\frac{\mathcal{F}(\mathbf{c}, t)}{u}\right), \quad \mathcal{F} = \text{Tr } \Phi_t(A(t)). \quad (\text{RC5})$$

XD4–7 supply the uniform contour estimates that justify holomorphy and differentiation. XD16–20 supply the nonzero monomial initial determinant and an adjugate proof valid even at a hypothetical singular matrix. Thus $\Pi(t)$ is invertible for every complex t , without deleting repeated-root fibres. The gamma multiplication formula simplifies XD18 to

$$\det(\Pi(t)^*\Pi(t)) = (2\pi|u|)^q \exp\left(2 \text{Re} \frac{\mathcal{F}(\mathbf{c}, t)}{u}\right). \quad (\text{RC6})$$

This last simplification is already in the supplied SGA continuation. It is not a new result of the present note. Bloch and Esnault (1999, §5) study period determinants of this polynomial-exponential type; the literal normalizations used here are the ones proved in XD1–26.

2 The deformation is the residue rank-one map

Define the original residue functional and its bilinear pairing by

$$\ell([P]) = [S^{q-1}] \text{rem}_\chi P, \quad \beta(x, y) = \ell(xy). \quad (\text{RC7})$$

In the power basis, $\beta(S^i, S^j)$ is zero for $i + j < q - 1$ and one for $i + j = q - 1$. Reversing its columns therefore produces a triangular matrix with diagonal one. Consequently

$$\det \beta = (-1)^{q(q-1)/2}, \quad \beta^b : E \xrightarrow{\sim} E^\vee. \quad (\text{RC8})$$

This proof inverts no discriminant. It is the perfect residue pairing in the supplied finite-flat SGA calculation. Its contraction with multiplication by χ' gives the separate, potentially degenerate trace pairing:

$$\text{Tr}_E M_P = \ell(\chi' P). \quad (\text{RC9})$$

For completeness, the divided difference $\mathcal{C}_\chi(X, Y) = (\chi(X) - \chi(Y))/(X - Y)$ is the coevaluation tensor for (RC8): $(\ell \otimes 1)((P \otimes 1)\mathcal{C}_\chi) = P$ and its image under multiplication is χ' . The finite-dimensional evaluation–coevaluation trace then proves (RC9). Neither (RC8) nor (RC9) identifies the bilinear residue form with the positive Hermitian source form G_N .

In the fixed remainder frame the perturbation in (RC4) is exactly

$$\boxed{\mathcal{R}(v) = \ell(v) 1_E, \quad \mathcal{R} = 1_E \otimes \ell.} \quad (\text{RC10})$$

It follows that the pair $(A, \mathcal{R})$ generates every coefficient endomorphism:

$$\boxed{\mathbb{C}\langle A, \mathcal{R} \rangle = \text{End}_{\mathbb{C}}(E).} \quad (\text{RC11})$$

Proof. The vectors $A^i 1_E$, $0 \leq i < q$, are the power basis. By (RC8), the functionals $\ell \circ A^j$, $0 \leq j < q$, are a basis of $E^\vee$. Therefore the q^2 operators

$$A^i \mathcal{R} A^j = (A^i 1_E) \otimes (\ell \circ A^j) \quad (\text{RC12})$$

are a vector-space basis of $\text{End}_{\mathbb{C}}(E)$. Every one belongs to the algebra generated by A and $\mathcal{R}$.

Now fix an actual nonzero proper invariant constituent, with its full maps,

$$0 \longrightarrow F \xrightarrow{\iota} E \xrightarrow{\pi} E/F \longrightarrow 0, \quad A\iota = \iota A_F, \quad 0 < p := \dim F < q. \quad (\text{RC13})$$

The deformation has a specified transverse map

$$\boxed{\pi \mathcal{R} \iota = (\pi 1_E) \otimes \ell_F : F \longrightarrow E/F, \quad \ell_F = \ell \circ \iota.} \quad (\text{RC14})$$

Both tensor factors are nonzero. If 1_E belonged to F , its entire A -orbit would belong to F , contradicting $F \neq E$. If $\ell_F = 0$, then for $v \in F$, invariance would give $\ell(A^j v) = 0$ for every j . Nondegeneracy of (RC8) would force every $v \in F$ to be zero. Thus

$$\boxed{\text{rank}(\pi \mathcal{R} \iota) = 1, \quad \ker(\pi \mathcal{R} \iota) = \ker \ell_F, \quad \text{im}(\pi \mathcal{R} \iota) = \mathbb{C} \pi 1_E.} \quad (\text{RC15})$$

In particular the kernel has dimension $p-1$. This is the strengthening of the supplied rank-at-most-one statement. It holds with all nilpotent jets retained. Equivalently, an invariant subspace is an ideal $(a)/(\chi)$ for a monic divisor $a \mid \chi$; the polynomial $S^{q-1-\deg a} a$ witnesses that ℓ_F is nonzero whenever $0 < \deg a < q$.

No nonzero proper *constant remainder-coordinate* subspace is invariant under $A(t)$ for every t . This assertion must not be called differential irreducibility: moving horizontal subbundles do exist. For example, $\Pi(t)^{-1} \Pi(0)F$ is horizontal. It is a different marked family from the fixed original constituent in (RC13).

3 Exact constituent-period curvature

Keep F and its inclusion fixed as in (RC13), and put

$$P_F(t) = \Pi(t)\iota, \quad H(t) = \Pi(t)^*\Pi(t), \quad H_F(t) = P_F(t)^*P_F(t) = \iota^*H(t)\iota. \quad (\text{RC16})$$

The target metric in this section is exactly the fixed coordinate metric I_q of the period construction. Since $\Pi(t)$ is invertible, $P_F(t)$ has rank p everywhere. Its orthogonal projector is

$$\mathbf{P}_F(t) = P_F(t)H_F(t)^{-1}P_F(t)^*. \quad (\text{RC17})$$

Use (RC4) and $A\iota = \iota A_F$ before projecting:

$$P'_F(t) = -\frac{1}{u}P_F(t)A_F - \frac{t}{u}\Pi(t)\mathcal{R}\iota. \quad (\text{RC18})$$

The first term is tangent to the same period subspace. Thus its complete normal derivative is

$$\boxed{\mathcal{N}_F(t) := (I - \mathbf{P}_F(t))P'_F(t) = -\frac{t}{u}(I - \mathbf{P}_F(t))\Pi(t)1_E\ell_F.} \quad (\text{RC19})$$

The column $(I - \mathbf{P}_F(t))\Pi(t)1_E$ never vanishes: vanishing would imply $1_E \in F$. Equations (RC15) and (RC19) prove that this normal map is zero at $t = 0$ and has rank exactly one at every $t \neq 0$. Its kernel for $t \neq 0$ is $\ker \ell_F$. At the marked fibre,

$$\boxed{(I - \mathbf{P}_F(0))P'_F(0) = 0, \quad (I - \mathbf{P}_F(0))P''_F(0) = -\frac{1}{u}(I - \mathbf{P}_F(0))\Pi(0)1_E\ell_F \neq 0.} \quad (\text{RC20})$$

The second identity follows by differentiating (RC5): $\Pi'' = \Pi A(t)^2/u^2 - \Pi\mathcal{R}/u$. At zero, $A^2\iota = \iota A_F^2$, so the A^2 term vanishes under the stated normal projection. The surviving map has rank one.

3.1 The exact coefficient of curvature

For any positive Hermitian matrix M on E , define

$$q_{M,F}([1_E]) = \min_{f \in F} \|1_E - \iota f\|_M^2, \quad M_F = \iota^*M\iota, \quad (\text{RC21})$$

and

$$\boxed{\mathfrak{c}_F(M) = q_{M,F}([1_E])\ell_F M_F^{-1}\ell_F^*.} \quad (\text{RC22})$$

This scalar is strictly positive for (RC13). It is independent of the basis chosen in F , but it retains the marked quotient vector $[1_E]$ and the marked residue functional ℓ_F . The two factors are the actual quotient norm and the actual dual norm, not arbitrary positive constants.

Let $\psi_F(t) = \log \det H_F(t)$, where the determinant is of the $p \times p$ positive Gram. No rectangular determinant is used. Ordinary matrix differentiation, using holomorphy of P_F , gives

$$\begin{aligned}\partial_t \partial_{\bar{t}} \psi_F &= \text{Tr} [H_F^{-1} P_F'^* P_F' - H_F^{-1} P_F'^* P_F H_F^{-1} P_F^* P_F'] \\ &= \text{Tr} [H_F^{-1} \mathcal{N}_F^* \mathcal{N}_F].\end{aligned}\tag{RC23}$$

The projector in (RC17) is self-adjoint and idempotent, which justifies the last equality. Further,

$$q_{H(t),F}([1_E]) = \|(I - P_F(t))\Pi(t)1_E\|^2.\tag{RC24}$$

Substitute (RC19) and take the trace of its rank-one product. This yields an exact identity for all complex t :

$$\partial_t \partial_{\bar{t}} \log \det H_F(t) = \frac{|t|^2}{|u|^2} \mathbf{c}_F(H(t)).\tag{RC25}$$

Thus the dual determinant line of the period-image subbundle, with its specified induced metric, has Chern form

$$c_1((\det \mathcal{F}_{\text{per}})^\vee) = \frac{i}{2\pi} \frac{|t|^2}{|u|^2} \mathbf{c}_F(H(t)) dt \wedge d\bar{t}.\tag{RC26}$$

It is strictly positive away from $t = 0$ and has quadratic vanishing there. Equivalently,

$$\partial_t^2 \partial_{\bar{t}}^2 \log \det H_F(t)|_{t=0} = \frac{\mathbf{c}_F(H(0))}{|u|^2} > 0.\tag{RC27}$$

The coefficient of $t^2 \bar{t}^2$ in $\log \det H_F$ is therefore $\mathbf{c}_F(H(0))/(4|u|^2)$; the factor four comes from $2!2!$. This mixed derivative is not the second derivative along the real axis. For $F = 0$ or $F = E$, the relevant curvature is identically zero; (RC13) is essential for strict positivity.

3.2 The first derivative still records the aggregate defect

At $t = 0$ and real $u > 0$, (RC18) gives

$$-u \frac{d}{dt} \log \frac{\det H_F(t)}{\det G_{N,F}} \Big|_{t=0} - kp = 2 \text{Re Tr } A_F - kp, \quad G_{N,F} = \iota^* G_N \iota.\tag{RC28}$$

This is the supplied constituent-minor observer, not a new estimate. Taking $F = E_>$, the complete positive-real-defect generalized eigenspace, gives the existing aggregate spectral defect. Equations (RC25)–(RC27) add the exact transverse curvature carried by that same marked constituent. A derivative of the full determinant alone sees only the balanced total trace on a reflection-stable packet.

4 Transfer to the original theta metric

4.1 A source-space expression for the coefficient

Using (RC2), the same coefficient for the actual arithmetic metric is

$$\mathfrak{c}_F(G_N) = \left(\min_{f \in F} \|r_N(1_E - \iota f)\|^2 \right) \left(\sup_{0 \neq f \in F} \frac{|\ell_F(f)|^2}{\|r_N \iota f\|^2} \right). \quad (\text{RC29})$$

The supremum equals the finite dual norm in (RC22). Every source norm, the arithmetic unit implicit in r_N , and every allowed theta correction remain in this expression.

The isometry from the original source metric into period coordinates is not $E \xrightarrow{\Pi} (\mathbb{C}^q, I_q)$. It is

$$(E, G_N) \xrightarrow{\Pi(t)} (\mathbb{C}^q, \widetilde{G}_N(t)), \quad \widetilde{G}_N(t) = \Pi(t)^{-*} G_N \Pi(t)^{-1}. \quad (\text{RC30})$$

Consequently the corresponding quotient-unit/residue product is invariant when the subspace, unit, functional, and metric are all transported: $F \mapsto \Pi F$, $1_E \mapsto \Pi 1_E$, and $\ell \mapsto \ell \Pi^{-1}$. In particular it equals $\mathfrak{c}_F(G_N)$, independently of t . This does not identify the Chern form for a varying target metric with the fixed-target-metric form (RC26).

4.2 A finite condition-number bridge

Let $a_N(t) > 0$ and $b_N(t) > 0$ be the least and greatest eigenvalues of

$$B_N^{\text{per}}(t) = G_N^{-1} H(t). \quad (\text{RC31})$$

This operator is positive and self-adjoint in G_N , so $a_N G_N \preceq H(t) \preceq b_N G_N$. Minimizing a quadratic form over a fixed coset preserves this order, whereas inversion of the restricted positive form reverses it. Therefore

$$\frac{a_N}{b_N} \mathfrak{c}_F(G_N) \leq \mathfrak{c}_F(H(t)) \leq \frac{b_N}{a_N} \mathfrak{c}_F(G_N). \quad (\text{RC32})$$

Writing $\mathcal{K}_N = b_N/a_N$, the exact curvature has the two-sided source-sensitive enclosure

$$\frac{|t|^2}{|u|^2 \mathcal{K}_N(t)} \mathfrak{c}_F(G_N) \leq \partial_t \partial_{\bar{t}} \log \det H_F(t) \leq \frac{|t|^2 \mathcal{K}_N(t)}{|u|^2} \mathfrak{c}_F(G_N). \quad (\text{RC33})$$

The calculated determinant of B_N^{per} determines its eigenvalue product, not $\mathcal{K}_N$. Equation (RC33) keeps the precise additional metric quantity required to compare the two curvature observations.

4.3 The existing endpoint certificate controls variation of this coefficient

For the original nested source degrees $i \leq j$, let $G_i \succeq G_j \succ 0$, $V_n = \det G_n$, and

$$\theta = \lambda_{\min}(G_i^{-1}G_j) \in (0, 1]. \quad (\text{RC34})$$

Then $\theta G_i \preceq G_j \preceq G_i$. The quotient norms satisfy $\theta q_{G_i, F} \leq q_{G_j, F} \leq q_{G_i, F}$, and the squared restricted dual norms satisfy $\|\ell_F\|_{G_i^{-1}}^2 \leq \|\ell_F\|_{G_j^{-1}}^2 \leq \theta^{-1} \|\ell_F\|_{G_i^{-1}}^2$, with each inverse taken on the restricted form. Multiplying gives

$$\theta \mathfrak{c}_F(G_i) \leq \mathfrak{c}_F(G_j) \leq \theta^{-1} \mathfrak{c}_F(G_i). \quad (\text{RC35})$$

Every eigenvalue of $G_i^{-1}G_j$ lies in $(0, 1]$, and one equals θ . Their product is at most θ . Hence

$$\left| \log \frac{\mathfrak{c}_F(G_j)}{\mathfrak{c}_F(G_i)} \right| \leq -\log \theta \leq \log \frac{V_i}{V_j}. \quad (\text{RC36})$$

This is a new use of the existing endpoint quantity. Any certified upper bound for the original $\log(V_i/V_j)$ immediately bounds the change of the specified residue/quotient coupling, without a separately assumed uniform spectral gap. It does not reverse into an upper bound for $\log(V_i/V_j)$ from the coupling alone.

5 The period realization of the arithmetic Laplacian

The quadratic polynomial of the original arithmetic generator is

$$\mathbf{L}_k = A^2 - kA. \quad (\text{RC37})$$

For $k = 1$ this is the polynomial corresponding to Connes–Consani’s $H(1 + H)$ under $H = -D$. A nonconstant parameter connection introduces an exact additional term at second order. From (RC5),

$$\Pi''(t) = \frac{1}{u^2} \Pi(t) A(t)^2 - \frac{1}{u} \Pi(t) \mathcal{R}. \quad (\text{RC38})$$

Thus, on every fibre,

$$u^2 \Pi''(t) + ku \Pi'(t) + u \Pi(t) \mathcal{R} = \Pi(t) (A(t)^2 - kA(t)). \quad (\text{RC39})$$

At zero this is the exact period comparison for (RC37). Omitting $u \Pi \mathcal{R}$ would give a false equation. The correction is the same residue map that supplies (RC15) and (RC25).

The original source lift of this correction is

$$\mathcal{R}_N^{\text{src}} = r_N \mathcal{R} j_E : \mathcal{X} \rightarrow \mathcal{X}, \quad j_E \mathcal{R}_N^{\text{src}} = \mathcal{R} j_E, \quad \mathcal{R}_N^{\text{src}} d = 0. \quad (\text{RC40})$$

It is defined on the specified source domain. For a fixed $x \in \mathcal{X}$ let $Y_x(t) = \Pi(t)j_E x$. Applying (RC39) at zero and using (RC2) gives

$$(u^2 \partial_t^2 + ku \partial_t)Y_x|_0 + u\Pi(0)j_E \mathcal{R}_N^{\text{src}} x = \Pi(0)j_E((D^{(k)})^2 - kD^{(k)})x. \quad (\text{RC41})$$

The original primitive for $D^{(k)}r_N - r_N A = dK_N$ is still present; the perturbed source operator $D^{(k)} + t\mathcal{R}_N^{\text{src}}$ has the same primitive, exactly as in DS69.

For the original Gram, retain

$$W_N = A^*G_N + G_N A - kG_N. \quad (\text{RC42})$$

Direct expansion proves

$$\mathbb{L}_k^* G_N - G_N \mathbb{L}_k = A^* W_N - W_N A, \quad G_N \mathbb{L}_k = W_N A - A^* G_N A. \quad (\text{RC43})$$

These are the preceding Laplacian-control identities, included here to fix the map to the present period calculation. Under (RC30), put $\tilde{A} = \Pi A \Pi^{-1}$, $\tilde{\mathbb{L}}_k = \Pi \mathbb{L}_k \Pi^{-1}$, and $\tilde{W}_N = \Pi^{-*} W_N \Pi^{-1}$. Substitution gives

$$\tilde{A}^* \tilde{G}_N + \tilde{G}_N \tilde{A} - k \tilde{G}_N = \tilde{W}_N, \quad (\text{RC44})$$

and the two identities (RC43) in precisely the transported metric. Thus every original numerical-range/control estimate is transported by an actual isometry, not by exchanging G_N for $H(t)$.

The full primary action is retained:

$$\mathbb{L}_k|_{E_\rho} = \rho(\rho - k)I + (2\rho - k)N_\rho + N_\rho^2. \quad (\text{RC45})$$

The finite residue algebra and the adelic Hochschild source are not being declared isomorphic. The latter reaches these finite coefficients through the original theta complex, its quotient, and j_E . In particular a theta boundary is killed by j_E ; the differentiated original relation producing the Jacobian trace is taken before that quotient. The finite algebra's higher Hochschild modules from the supplied periodic resolution, $HH_{2a+1} = E/(\chi')$ and $HH_{2a+2} = \text{ann}_E(\chi')$, remain separate from the operator $\mathcal{R}$ and from the adelic Hochschild groups.

6 Translation, cohomological support, and exact scope

6.1 Translation preserves curvature but changes the marked trace

The supplied DT1–13 comparison is

$$\chi_a(S) = \chi(S - a), \quad \Phi_a(S) = \Phi(S - a) - \Phi(-a), \quad \Pi_a(t) = f_a(t)\Pi(t)C_a, \quad f_a(t) = e^{(-\Phi(-a)-ta)/u}, \quad (\text{RC46})$$

where C_a is the Pascal substitution matrix, $\det C_a = 1$, and $C_a A_a C_a^{-1} = A + aI$. Transport the same constituent by $\iota_a = C_a^{-1} \iota$. Then

$$P_{F,a}(t) = f_a(t) P_F(t), \quad \det H_{F,a}(t) = |f_a(t)|^{2p} \det H_F(t). \quad (\text{RC47})$$

Since f_a is nowhere zero and holomorphic,

$$\boxed{\partial_t \partial_{\bar{t}} \log \det H_{F,a}(t) = \partial_t \partial_{\bar{t}} \log \det H_F(t).} \quad (\text{RC48})$$

But, with the original k fixed,

$$2 \operatorname{Re} \operatorname{Tr}(A_F + aI) - kp = (2 \operatorname{Re} \operatorname{Tr} A_F - kp) + 2p \operatorname{Re} a. \quad (\text{RC49})$$

Thus curvature, unlike the first-derivative observer, does not retain this scalar normalization by itself. The exact scalar f_a and the marked arithmetic fibre $a = 0$ remain necessary parts of the comparison. This is not an assertion that translated polynomials are actual zeta packets. In the specified finite-field realization DT12–13 instead tensors with the constant Artin–Schreier character line, whose geometric-Frobenius scalar is a root of unity. The finite-field weights are preserved by that operation; it does not identify that Frobenius with A .

6.2 Cohomological degrees and supported-zero maps

For an invariant F , the supplied logarithmic lattice $\mathcal{M}_F = F\mathcal{O} + E\mathcal{O}(-D)$ gives

$$\mathbb{H}^0(\mathcal{K}_F) = F, \quad \mathbb{H}^1(\mathcal{K}_F) = 0, \quad \mathbb{H}^2(\mathcal{K}_F) = (E/F) \otimes \mathbb{T}. \quad (\text{RC50})$$

Its support-compatible dual is $\mathcal{M}_F^\vee(-D) = F^{\text{ann}}\mathcal{O} + E^\vee\mathcal{O}(-D)$. The evaluation target includes that divisor twist. The full coefficient extension $0 \rightarrow F \rightarrow E \rightarrow E/F \rightarrow 0$ is retained through the outer comparison complexes; it is not reconstructed from the two middle cohomology groups alone. These are the corrected distinctions in DS25–34 and DS66–69.

The new transverse map (RC14) is therefore a specified map $\mathbb{H}^0(\mathcal{K}_F) \rightarrow E/F$, with the latter connected to degree two by the retained line $\mathbb{T}$. It is not silently a degree-zero chain endomorphism of $\mathcal{K}_F$. In particular, $\mathcal{R}$ need not preserve its lattice; (RC15) quantifies the failure.

Lift every coefficient map by the original support reconstruction, retaining the structural square

$$\begin{array}{ccc} G(\mathbb{Z}) & \longrightarrow & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \longrightarrow & \mathbb{C}. \end{array} \quad (\text{RC51})$$

For each support label λ , the new map is

$$(\lambda, v) \mapsto (\lambda, (\pi 1_E) \ell_F(v)), \quad \tau \mapsto \tau. \quad (\text{RC52})$$

If $v \in \ker \ell_F$, its receiving value is the supported zero at λ , not external absence. For the single active fibre,

$$S(f)^{-1}(\tau) = \{\tau\}, \quad S(f)^{-1}(e) = (\ker f)^\bullet, \quad p^{-1}(\ker f) = \{\tau\} \sqcup (\ker f)^\bullet. \quad (\text{RC53})$$

This uses the corrected split-kernel formula in the supplied review. The original infinite arithmetic quotient, support transitions, and all source boundaries remain attached.

7 Verification and provenance

The included checker contains twelve independent finite test methods. Its fixtures are monic polynomials with rational coefficients, including repeated roots, and exact complex-rational matrices. It checks the residue pairing and trace, the q^2 endomorphism basis, the exact transverse rank and kernel, the curvature factorization, a bivariate fourth-order log-Gram jet with its factorial, the metric enclosures, the forced Laplacian equation, and the constituent translation comparison. The normal and optimized runs have their actual command and output records in `checks/`. Three deliberately false formulas are tested separately: zero transverse rank, omitted Laplacian forcing, and omitted $|t|^2$ in the curvature. These must fail in both modes.

The finite tests do not prove improper-integral convergence, the global analytic period theorem, Deligne’s theorem, or uniform growth on arithmetic packets. Those parts have their written source proofs and stated hypotheses. No predecessor checker or Lean build is claimed as a new execution here. No remote repository or publication is modified by this package.

The dated *Deligne Primary Source Correction* supersedes the historical unresolved-citation claim in *Exponential Intake*: the reference is Roman (I.8.11) to Weil I, printed page 306. The strict weight inequality, the tensor square of the same eigenvalue, and the lisse/mixed hypotheses are retained. This note relies on that supplied correction for those printed-page identifications; it does not claim a fresh full audit of Weil II.

The intake receipt records the hashes actually measured from the uploaded files. All nine files listed in the manifest’s `copies` array match their output hashes. The currently supplied README is newer than the README hash recorded in that manifest. The independent review also pins translation TeX hash 39163624..., whereas the delivered translation source and the manifest agree on bd1831cd.... The review’s source-byte identity must therefore be reconciled before its acceptance is attributed to those exact delivered translation bytes; this discrepancy does not itself show that a displayed mathematical formula is false.

8 References and supplied-source locators

Bloch, S., & Esnault, H. (1999). *Gauß–Manin determinant connections and periods for irregular connections* [Preprint]. arXiv, math/9912095. The inspected §5 supplies the polynomial-exponential period context, not the new assertions (RC11), (RC15), (RC25), or (RC36).

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and ζ -cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83–101). American Mathematical Society. DOI: 10.1090/pspum/105/01896.

Deligne, P. (1980). La conjecture de Weil: II. *Publications Mathématiques de l’IHÉS*, 52, 137–252. DOI: 10.1007/BF02684780. The present note does not infer characteristic-zero spectral purity from finite-field purity.

National Institute of Standards and Technology. (n.d.). §5.5: Functional relations. In *Digital Library of Mathematical Functions*. Equation 5.5.6, Gauss’s gamma multiplication formula, gives the constant in (RC6).

Deligne Split Cohomology. (2026, September 13). Supplied research sidebar, DS1–69. Source: `deligne_split_sidebar_20260913.tex`; relevant maps DS12–14, DS25–34, DS53–69.

Exponential Period Determinant. (2026, September 13). Supplied research note, XD1–26. Source: `deligne_exponential_determinant_extension_20260913.tex`.

Exponential Translation Comparison. (2026, September 13). Supplied research note, DT1–13. Source: `Exponential_Translation_Comparison.tex`.

Pasted markdown (20260913-034115). (2026, September 13). Supplied SGA trace/period continuation. Sections 2–3 give the residue coevaluation and periodic resolution; §§4–5 evaluate and differentiate the determinant; §§6–7 identify the constituent minor and transport the original metric.

Independent Deligne Review, Exponential Intake, and Deligne Primary Source Correction. (2026, September 13). Supplied review and correction records. Their historical execution and source-reading claims are not relabeled as new executions in this note.